

Project Proposal

Learning to See in the Dark

*Based on: Chen et al., CVPR 2018 | Course: Foundation of Computer Vision | Student: Talha Asim
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Group Members

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Problem Statement

Capturing images in extremely low-light conditions is a fundamental challenge in computer vision. Traditional image processing pipelines (demosaicing, denoising, white-balance) amplify noise and lose detail when applied to dark raw images. This project aims to reproduce and study the end-to-end deep learning pipeline from Chen et al. (CVPR 2018) that directly maps short-exposure raw sensor data to a clean, well-exposed RGB image — bypassing the conventional ISP entirely.

Proposed Approach

We implement the fully-convolutional U-Net pipeline proposed by Chen et al., which operates directly on packed raw Bayer data. The key stages are:

1. Pack the 4-channel Bayer mosaic (RGGB) and amplify by the exposure ratio ($\times 100$ – $\times 300$)
2. Feed into a U-Net encoder–decoder with skip connections that preserves spatial detail
3. Upsample the final feature map to full RGB resolution (sub-pixel convolution / depth-to-space)
4. Train end-to-end with L1 loss against long-exposure reference RGB images
5. Evaluate on held-out scenes with PSNR, SSIM, and visual inspection

Key design choices to explore:

- **Amplification ratio:** Scale raw values by the GT exposure ratio before the network — simulates longer exposure
- **Loss function:** Compare L1 vs. perceptual (VGG) loss for colour fidelity and texture sharpness
- **Architecture depth:** Ablate number of U-Net levels (3 vs. 5) to study the quality–speed trade-off

Dataset & Tools

- **Dataset:** See-in-the-Dark (SID) — 5,094 raw/long-exposure pairs captured with Sony $\alpha 7S$ II and Fujifilm X-T2 (~25 GB, publicly available)
- **Input format:** Raw .ARW/.RAF files unpacked to 4-channel Bayer tensors via rawpy
- **Framework:** Python 3.10, PyTorch 2.x, rawpy, NumPy, scikit-image, Matplotlib
- **Metrics:** PSNR and SSIM on the Sony test split; qualitative side-by-side comparisons

Timeline

- **Week 1:** Dataset download, raw file parsing with rawpy, data loader & augmentation pipeline
- **Week 2:** U-Net implementation, exposure-ratio amplification, L1 training baseline
- **Week 3:** Loss ablations (L1 vs. perceptual), architecture depth experiments
- **Week 4:** Quantitative evaluation (PSNR / SSIM), qualitative visualisations

- **Week 5:** Final report, demo notebook, submission

Expected Outcomes

We expect to reproduce the paper's core result — visually clean, colour-accurate images from very dark raw captures — and to demonstrate the advantage of end-to-end raw-domain learning over traditional ISP pipelines. Deliverables include: a trained U-Net model, PSNR/SSIM evaluation tables, side-by-side low-light vs. enhanced image comparisons, and a final report.

Reference: C. Chen, Q. Chen, J. Xu, V. Koltun. "Learning to See in the Dark." CVPR 2018.